

Tic-Tac-Toe

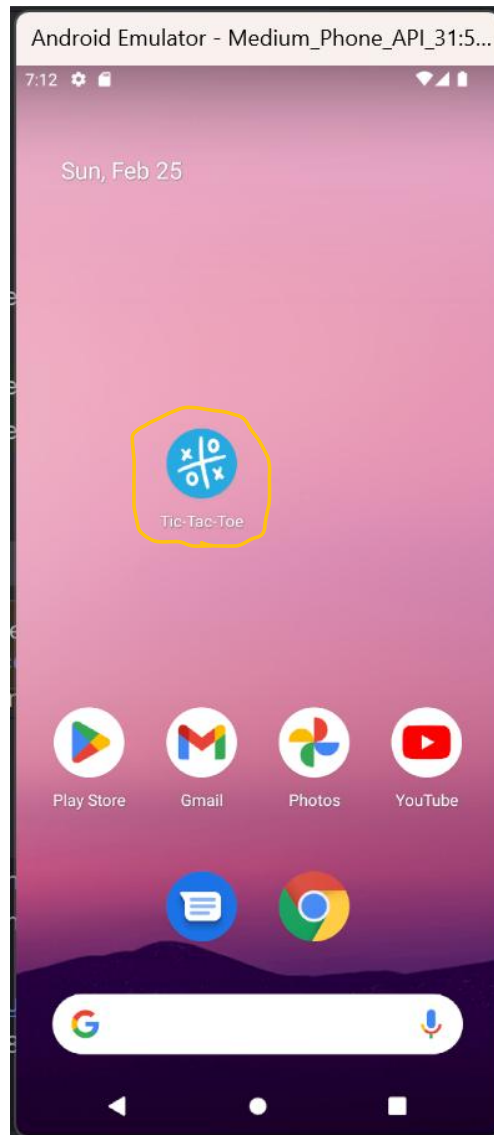


Figure 1: Icon of the app

The application icon was redesigned from the default Android icon to a Tic-Tac-Toe icon, allowing users to recognise the game's purpose immediately and reducing any potential confusion when locating the application.

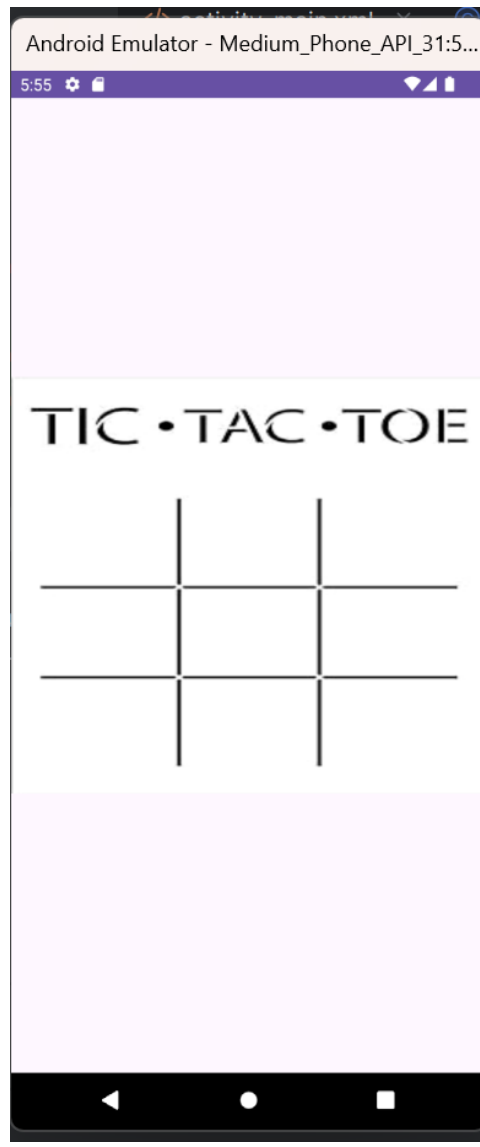


Figure 2: After opening the app

Figure 2 shows the main game interface after the user launches the application from Figure 1. A white background with black grid lines and text was selected to provide a clean layout and improve visibility, ensuring that the game interface is easy to read and use.



Figure 3: 1st player starts the game.

The game begins when the first player selects any one of the nine available grid boxes. The selected box is then marked with the player's symbol.



Figure 4: 2nd player continues.

The second player takes their turn by selecting one of the remaining eight empty grid boxes. Once a box has been selected, it cannot be chosen again for the remainder of the game.

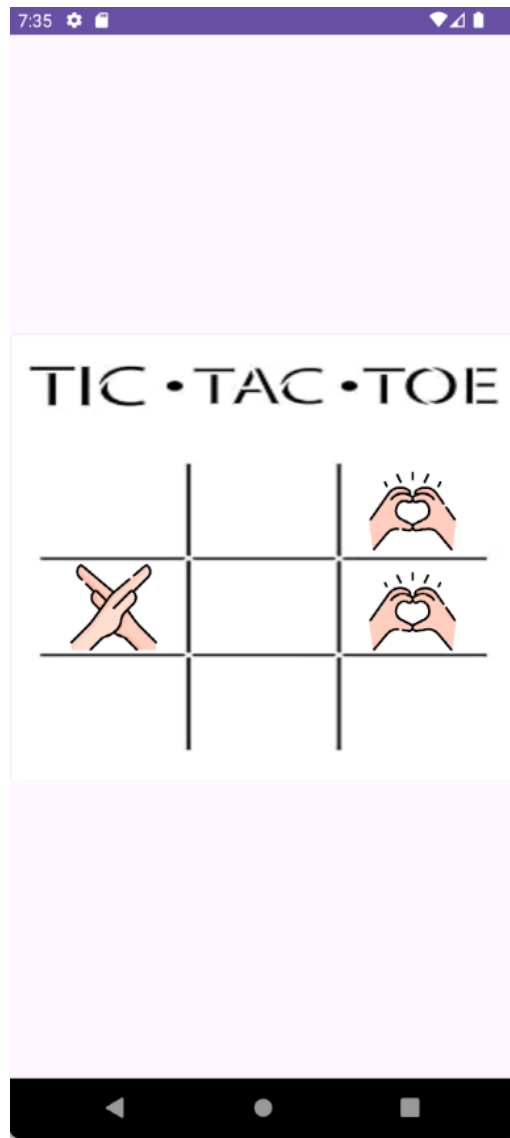


Figure 5: 1st player continues.

The players continue taking turns by selecting the remaining empty grid boxes. The game proceeds until either player forms a continuous line of three identical symbols horizontally, vertically, or diagonally. If no winning combination is achieved after all nine boxes have been filled, the game ends in a draw.

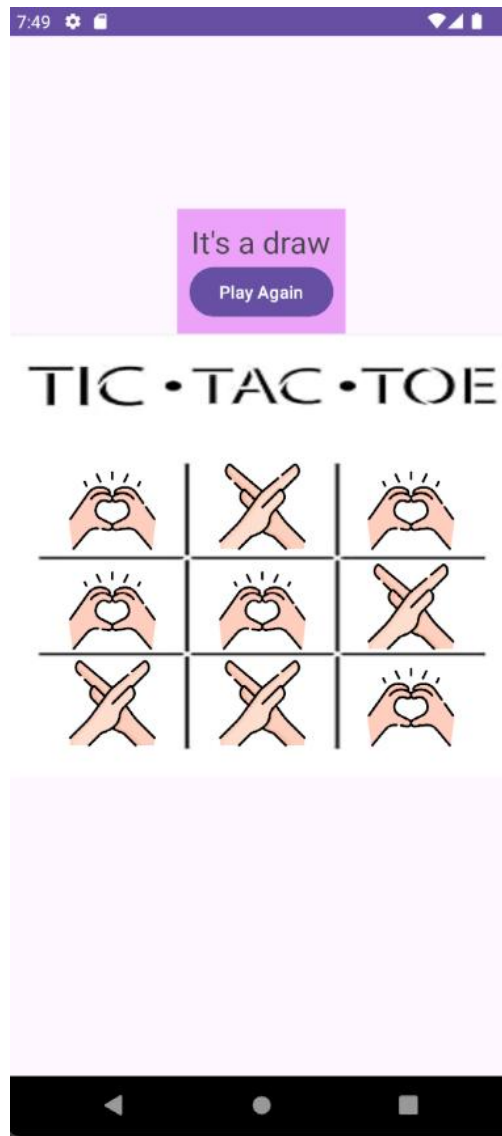


Figure 6: Draw

If neither player forms a line of three identical symbols after all nine grid boxes have been selected, the game is declared a draw. The user may then press the **"Play Again"** button to restart the game, which resets the board to its initial state as shown in Figure 2.

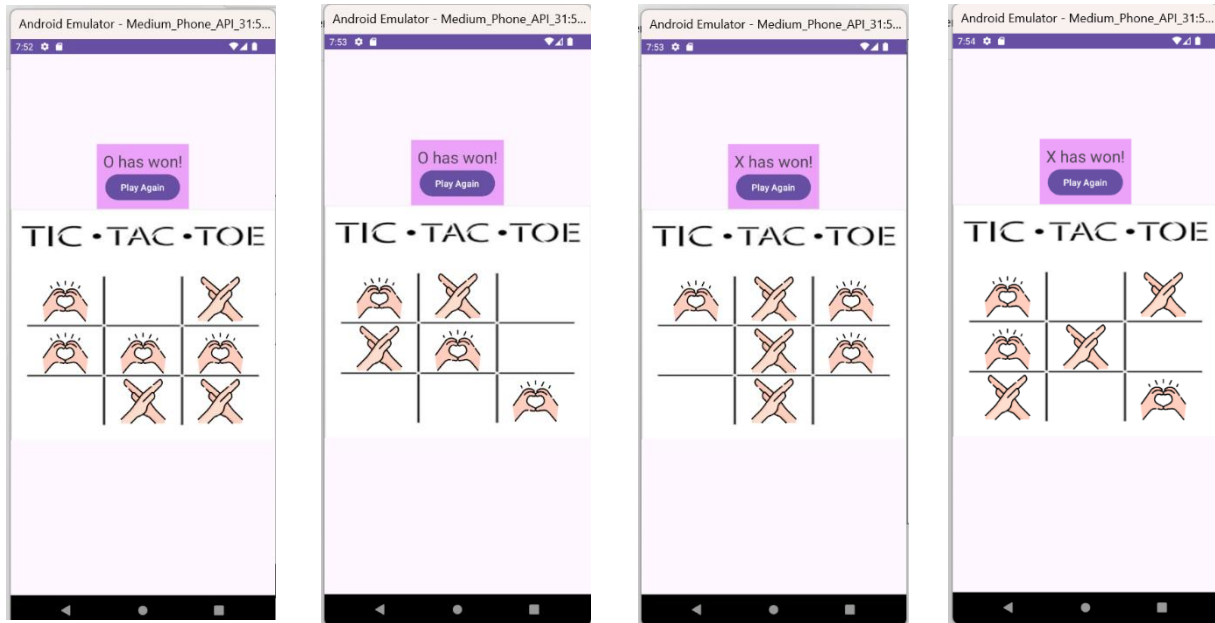


Figure 7, 8, 9, 10: 1 of the player wins.

When either player successfully forms a line of three identical symbols, a winning screen is displayed. Figures 7 and 8 illustrate the outcome when Player **O** wins, while Figures 9 and 10 show the outcome when Player **X** wins. Users can press the "**Play Again**" button to restart the game, returning the application to the initial game interface shown in Figure 2.

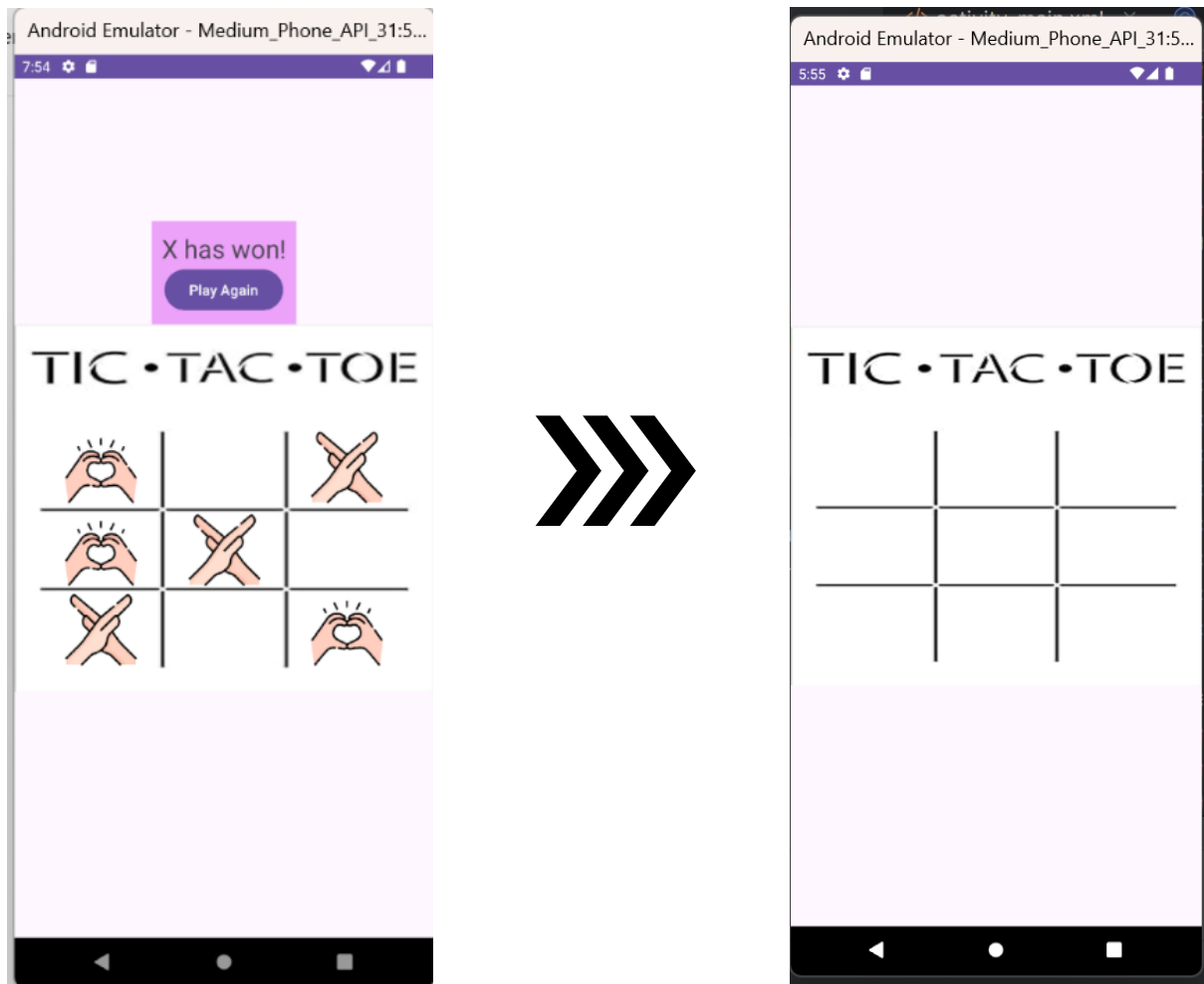


Figure 11: Play again

When the **"Play Again"** button is selected, the Tic-Tac-Toe board is reset by clearing all previously selected grid boxes, allowing players to begin a new game.

Self-Evaluation Report

This project provided valuable hands-on experience in developing a mobile game using Android Studio. Throughout the development process, I gained a better understanding of Android Studio's features, the application development workflow, and the implementation of game logic. Completing this project also increased my interest in Android application development and strengthened my programming skills.

At the beginning of the semester, I found Android development challenging and often felt uncertain about how to approach the tasks. However, after completing this assignment, I became more confident in applying the concepts that I had learned. Although I have made significant progress, I recognise that there is still considerable room for improvement in both my technical skills and programming knowledge. If I were to evaluate my current ability on a scale of 1 to 10, I would rate myself a **5**, as I believe I still have much to learn for future assignments and projects.

During the development process, I encountered several coding errors and technical challenges. To overcome these difficulties, I referred to online resources such as YouTube tutorials. When these resources were insufficient, I sought guidance from my lecturer, who helped me identify my mistakes and provided valuable suggestions for improving my code and problem-solving approach.

Overall, this assignment was a rewarding learning experience. It enhanced my understanding of Android application development, improved my problem-solving abilities, and increased my confidence in tackling programming challenges independently. Moving forward, I intend to continue expanding my knowledge through self-learning and practical experience to further improve my software development skills.